Options

Menu: File > Options

The page has two tabs: **Speed** for the speed source and display settings, and **Phone** for connecting a phone to RateController's phone page.

Speed Tab

Selects the source of ground speed used for rate calculations and configures display units.

Speed Source

Option	Description
GPS	Use speed from AgOpenGPS
Simulated Speed	Simulate a constant speed (useful for testing)
Wheel Sensor	Use a speed sensor connected to a module GPIO pin

Wheel Sensor Setup

1. Enter the **Module ID** and **GPIO pin** used for speed sensing.
2. Enter the calibration number in **pulses per mile** or **pulses per kilometer**.
3. Press the **up arrow** to send configuration to the module (also sent on Save).
4. Use the **Calibration screen** to calculate the calibration number if unknown.

Display Options

Option	Description
Metric	Toggle between metric (km, ha, L) and imperial (mi, ac, gal) units
Compact Display	When on, the arrow on the main screen collapses it to just the rates instead of minimizing it to the small restore window. Tap the collapsed display to expand it again.

The rate display window is no longer switched on here. Each product now has its own **Popup Display** button on Products > Settings, so a separate window can be shown for each product.

Buttons

Button	Action
up arrow	Send wheel sensor configuration to the module
Calibration screen	Open the speed calibration screen
Save	Save settings (both tabs)
Cancel	Discard changes

Phone Tab

RateController has a built-in page for a phone's web browser, so the switches and calibration can be used away from the tablet: walking the boom to check nozzles, or standing at the meter with a catch bag. It is always available; there is nothing to install on the phone and nothing to switch on in RateController.

Connecting a Phone

1. Put the phone and the tablet on the **same network**. Any of these works:
    - **The phone's hotspot**: connect the tablet to it.
    - **A WiFi router** (Starlink, a cellular router or any other): connect both to its main network.
    - **The tablet's own hotspot**: choose **Tablet Hotspot** on Modules > Communication > WiFi and press Save, then connect the phone using the network name and password shown on this tab.
2. **Phone hotspot or router only:** once the tablet is connected, tap **WiFi settings**, tap the network and set it to **Private**. Windows blocks the phone page on a Public network. Do this again for a new phone or after renaming the phone's hotspot, because Windows treats it as a new network.
3. Scan the code with the phone's camera to open the page. Adding it to the phone's home screen makes it quicker next time.

Controls

Control	Description
QR code	Opens the phone page when scanned. When the tablet is on more than one network, tap the code to show the next address. It updates by itself when a network is joined or left.
Address	The same address as text, for typing into the phone's browser. "(1 of 3, tap for next)" shows when there are several.
Tablet Hotspot	The hotspot's state: On, Off, Checking..., *On, firewall not allowed* or *Off, restarts failed*. The hotspot is switched on and off on Modules > Communication > WiFi; the **Comm → WiFi** link opens that page.
Network / Password	The tablet hotspot's network name and password, for joining it from the phone, or Windows' reason when the hotspot can't start.
WiFi settings	Opens the Windows WiFi page, for setting a network to Private.

Tablet Hotspot

Set on Modules > Communication > WiFi. The same hotspot carries both the phone and any ESP32 modules joined to it.

- The first time Tablet Hotspot is saved, Windows asks for **administrator permission** to let the phone page through the firewall. This is needed once per tablet. If it's declined, the phone can join the hotspot but the page won't open, and the status shows *On, firewall not allowed*. On a standard Windows account the prompt asks for an administrator's password.
- The tablet's hotspot address is always **192.168.137.1**, so a bookmark or home-screen link on the phone keeps working.
- If the hotspot turns off, RateController turns it back on within about 30 seconds, up to 3 times. After that the status shows *Off, restarts failed*. To try again, choose Current WiFi Network on Comm → WiFi and Save, then Tablet Hotspot and Save, or restart RateController.

- Windows' **Power saving** turns the hotspot off when no device has been connected for 5 minutes. If necessary, turn it off in Windows' **Hotspot settings** (linked from Comm → WiFi). RateController can't change this setting.
- Windows shares the tablet's internet connection through the hotspot. If the tablet connects to a router by WiFi as well, its WiFi adapter must be able to do both at once; not all adapters can.

Router Tips

- Use the router's main network, not a guest network, and make sure **AP isolation / client isolation** is off. Either one stops the phone from reaching the tablet.
- A **DHCP reservation** for the tablet in the router keeps its address the same, so the phone's bookmark keeps working.
- The phone has to stay within the router's WiFi range at the boom.

Phone Page

The phone opens a menu titled **RateController** with two buttons:

- **Switches**: a copy of the on-screen switch panel. See Machine > Switches.
- **Calibration**: product calibration. See Machine > Calibrate.

Each page has a **Menu** button at the bottom to go back. The page uses the tablet's display colours.

> **Note:** The phone page has no password. Anyone on the same network can open it, so use your own hotspot or network.